

Task 4: Deploy a Docker Container on Cloud VM

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Objective

The objective of this task is to deploy a Docker-based web application on an AWS EC2 virtual machine and make it accessible over the internet using a public IP address and port mapping.

Tools Used

- AWS EC2
- Amazon Linux 2023
- Docker Engine
- Nginx Docker Image
- AWS Security Groups

Step 1: Verify Docker Installation

Docker was verified on the EC2 instance using the following command:

```
docker --version
```

Output:

Docker version 25.0.14

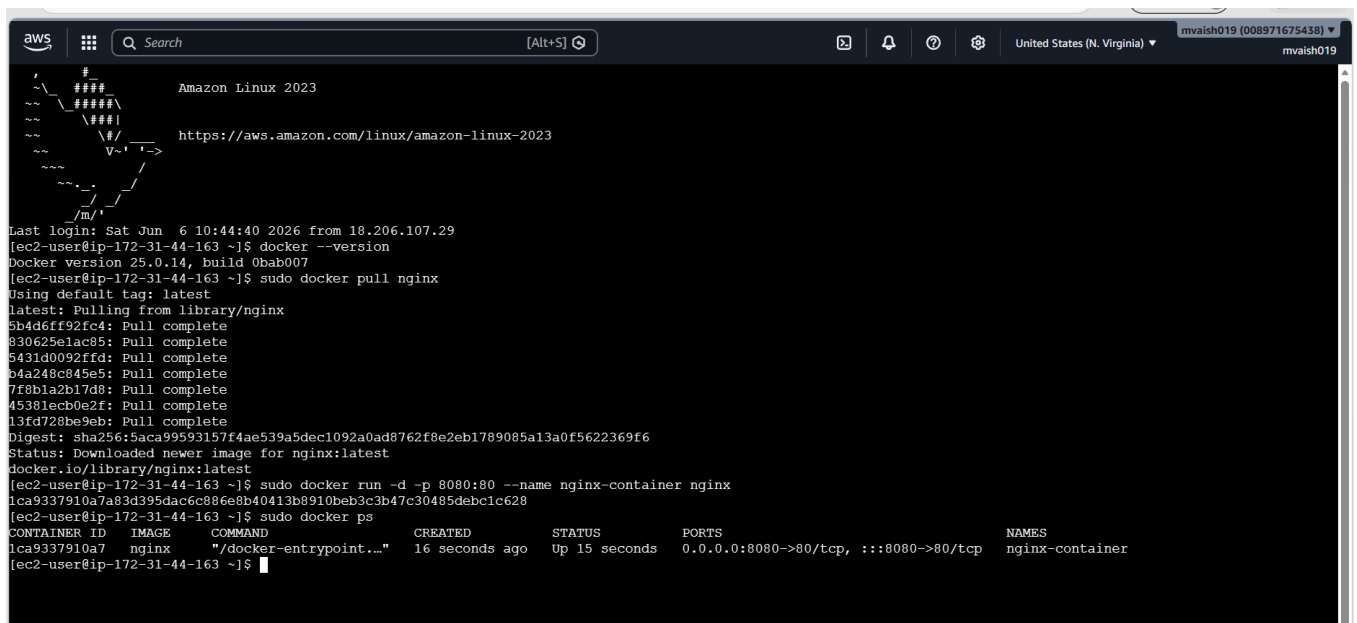
Step 2: Download Nginx Docker Image

The Nginx image was downloaded from Docker Hub.

Command:

```
sudo docker pull nginx
```

This command downloaded the latest Nginx image to the EC2 instance.



```
aws
Search [Alt+S]
United States (N. Virginia) mvalsh019 (008971675438) mvalsh019

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Sat Jun 6 12:02:04 2026 from 18.206.107.27
[ec2-user@ip-172-31-44-163 ~]$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS                               NAMES
1ca9337910a7   nginx    "/docker-entrypoint..." 3 minutes ago Up 3 minutes 0.0.0.0:8080->80/tcp, :::8080->80/tcp nginx-container
[ec2-user@ip-172-31-44-163 ~]$
```

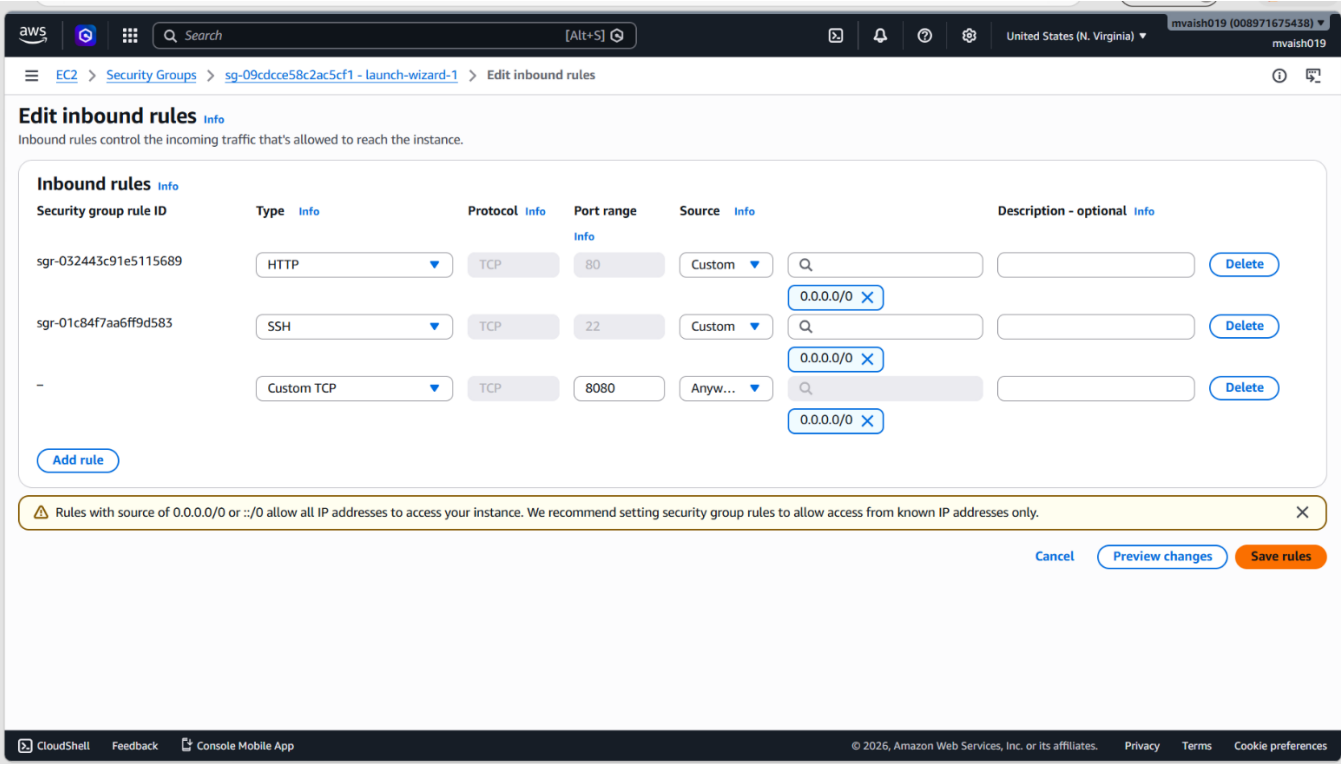
Step 5: Configure Security Group

To allow external access to the containerized application, a new inbound rule was added.

Configuration:

Type	Protocol	Port	Source
Custom TCP	TCP	8080	0.0.0.0/0

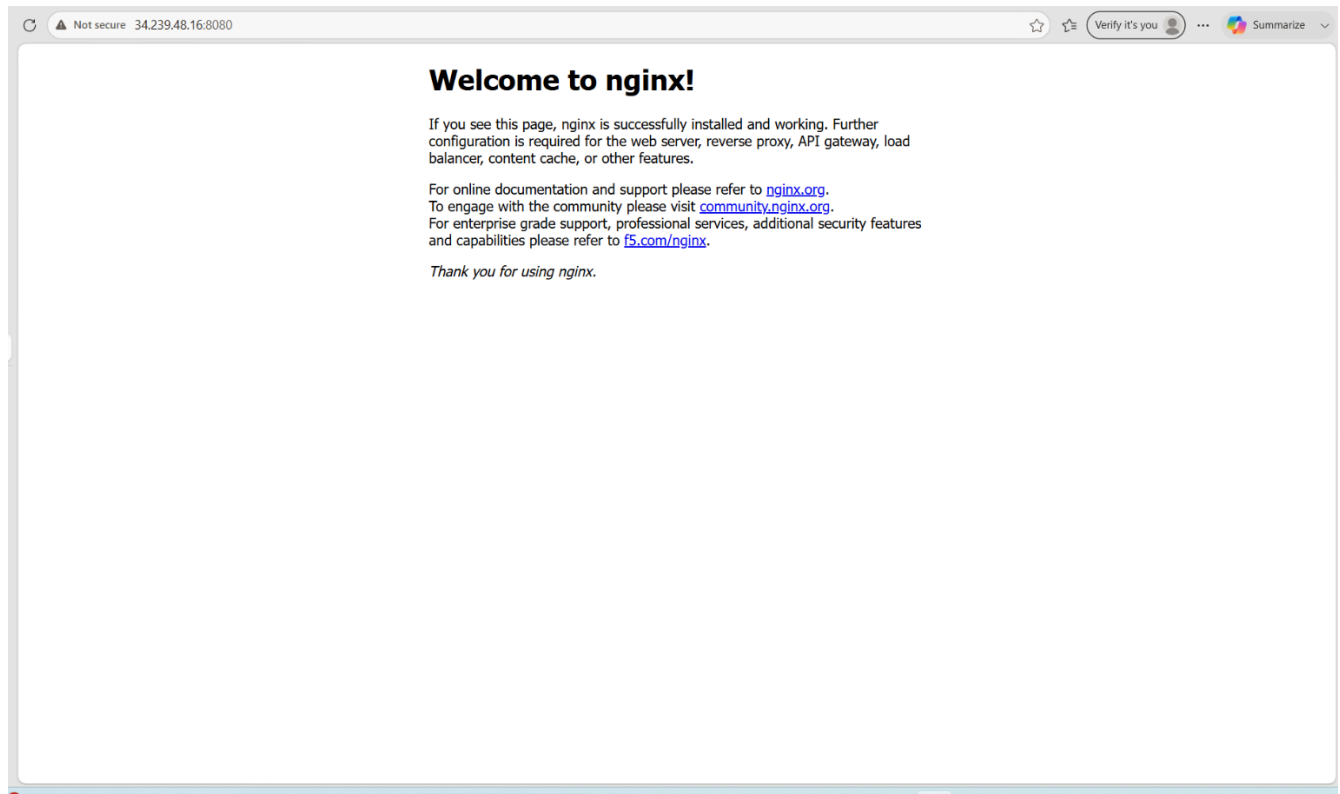
This allowed public access to the Docker container through port 8080.



Step 6: Access Application Through Public IP

The application was accessed using:

Welcome to nginx!



Result

The Docker-based Nginx web application was successfully deployed on an AWS EC2 virtual machine. The container was exposed to the internet using port mapping and AWS Security Group configuration, and the application was successfully accessed through the public IP address.

Conclusion

This task provided practical experience in containerization using Docker. It demonstrated how to deploy, manage, and expose containerized applications on a cloud virtual machine. The deployment was successfully completed using AWS EC2 and Docker, making the Nginx web server publicly accessible.